

Lab Title:- INVESTIGATING HARASSMENT EMAIL TRAFFIC WITH WIRESHARK

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Course Name:- COMPUTER FORENSICS CASE STUDIES

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Date Of Submission:- 05/10/2026

Case scenario

Lily Tuckrige, a Chemistry Department teacher, reports repeated harassing emails sent to her personal Yahoo Mail account. Full headers from an earlier message point to an IP address in a shared student residence where an open wireless router is in use. On Monday 21 July, a later message arrived through the web service willselfdestruct.com. The service briefly displayed the message and then reported that it had been destroyed. The university captured traffic at the residence network boundary and asks you to determine whether the available evidence supports attribution to a student in Chemistry 109.

Lab Objectives

The objectives of the lab is to understand how to use wireshark to analyse a pcap file/capture in order obtain evidences.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- wireshark:- is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
 - sha256sum:- Print/compute or check SHA256 (256-bit) checksums message digest.
 - Md5sum:- Print/compute or check MD5 (128-bit) checksums message digest.
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Methology

- I power on the kali linux machine
- I download the pcap capture(nitroba.pcap) and then make a copy of it

- I obtain the hash value of the pcap capture(nitroba.pcap)
- I upload the pcap capture(nitroba.pcap) into the wireshark and then start the analysis.

Screenshot and Evidence

SCREENSHOT SHOWING THE HASH VALUE(MD5/SHA256) OF THE PCAP FILE

```
(kali㉿kali)-[~]
└─$ cd CASE\ STCY\ 1

(kali㉿kali)-[~/CASE STCY 1]
└─$ ls
nitroba.pcap

(kali㉿kali)-[~/CASE STCY 1]
└─$ md5sum nitroba.pcap
9981827f11968773ff815e39f5458ec8  nitroba.pcap

(kali㉿kali)-[~/CASE STCY 1]
└─$ sha256sum nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb  nitroba.pcap

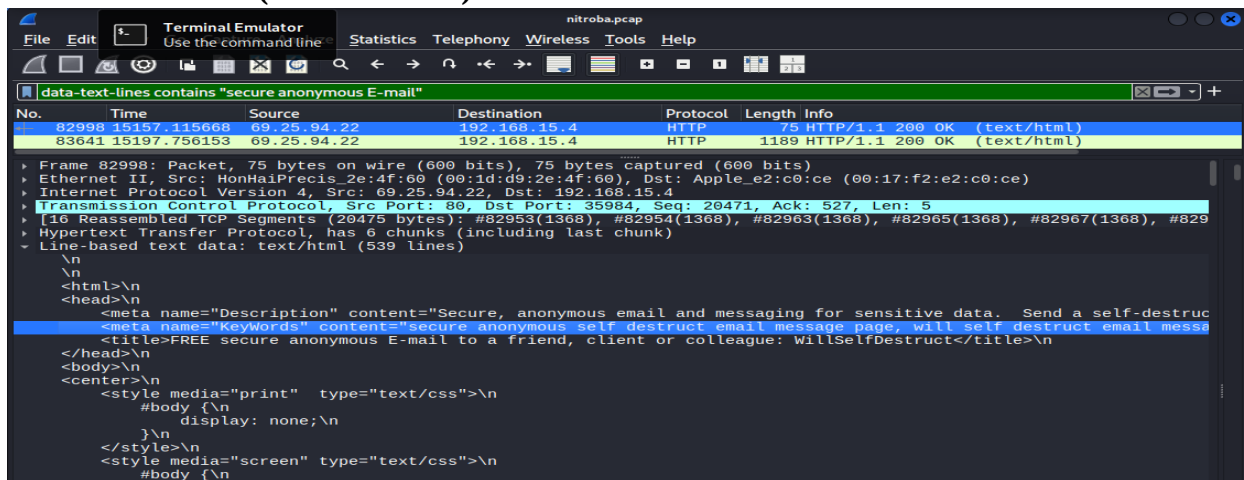
(kali㉿kali)-[~/CASE STCY 1]
```

SCREENSHOT 1 (EVIDENCE 1)

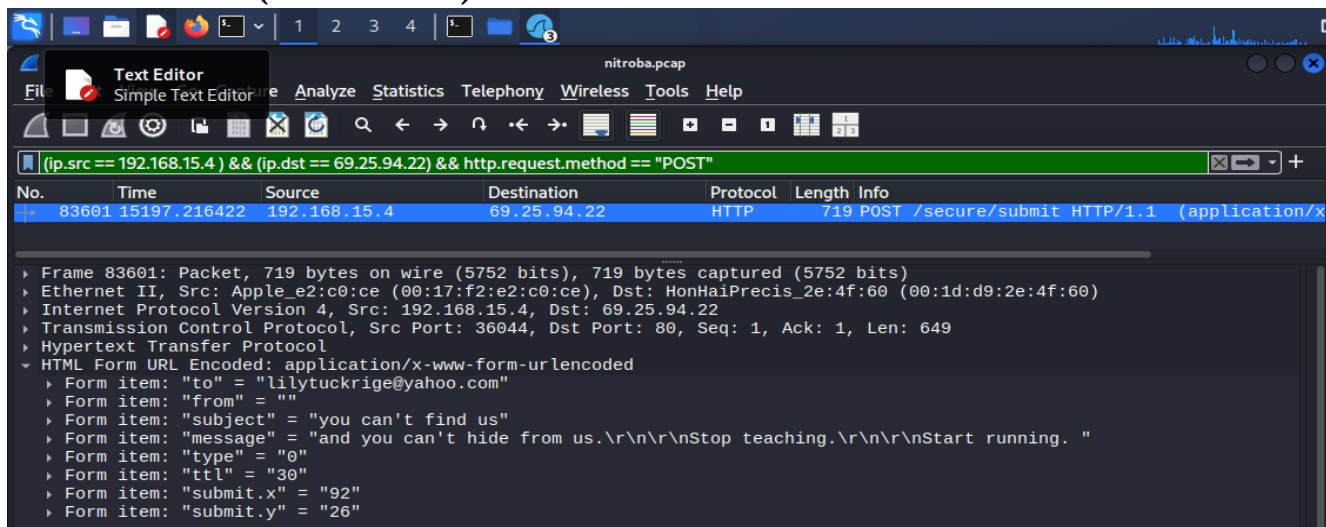
The screenshot shows the Wireshark interface with a packet list and packet details pane. The packet list shows three HTTP GET requests. The selected packet (No. 82936) is an HTTP GET request to /secure/submit. The packet details pane shows the following information:

- Frame 82936: Packet, 596 bytes on wire (4768 bits), 596 bytes captured (4768 bits)
- Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
- Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
- Type: IPv4 (0x0800)
- [Stream index: 19]
- Frame check sequence: 0x59d1a443 [unverified]
- [FCS Status: Unverified]
- Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
- Transmission Control Protocol, Src Port: 35984, Dst Port: 80, Seq: 1, Ack: 1, Len: 526
- Hypertext Transfer Protocol
 - GET /secure/submit HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /secure/submit
 - Request Version: HTTP/1.1
 - Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, Referer: http://email.about.com/gi/dynamic/offsite.htm?zi=1/XJ&sdn=email&cdn=compute&tm=178
 - Accept-Language: en-us\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
 - Host: www.willselfdestruct.com\r\n
 - Connection: Keep-Alive\r\n
 - \r\n
 - [Response in frame: 82998]

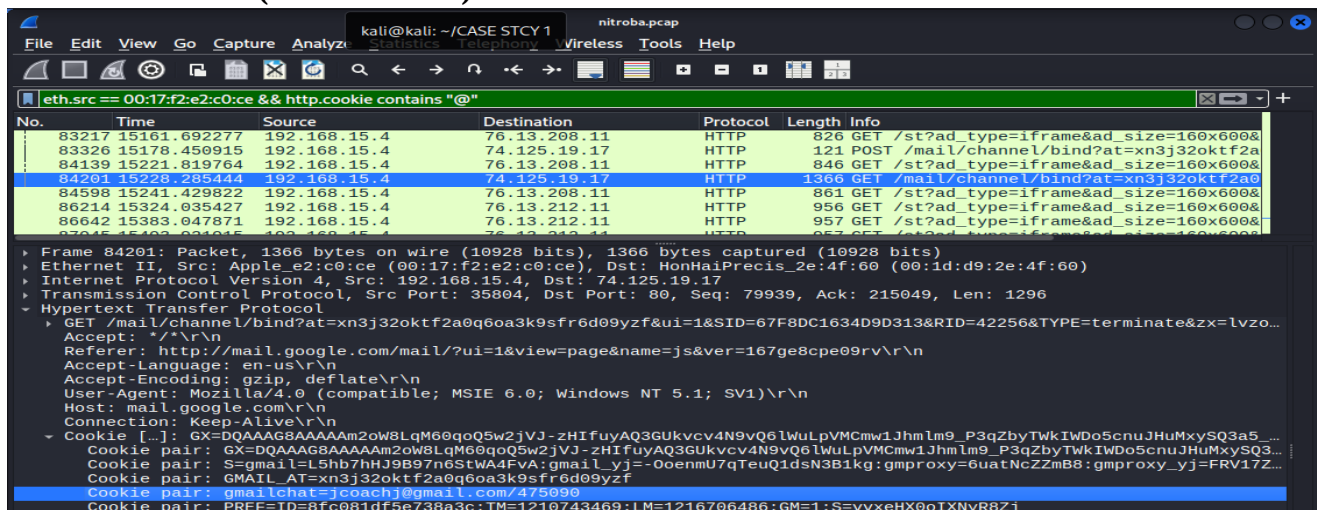
SCREENSHOT 2 (EVIDENCE 2)



SCREENSHOT 3 (EVIDENCE 3)



SCREENSHOT 4 (EVIDENCE 4)



EVIDENCE TEMPLATE

EVIDENCE ID	PACKET/ITEM	FINDINGS	WHY ITS MATTERS	SCREENS HOT NUMBER
E1	82936	A user with an ip address(192.168.15.4) and MAC Address(00:17:f2:e2:c0:ce) visited the website(www.willselfdestruct.com)which has an ip address(69.25.94.22)	It matters because it help me in identifying the ip address that visited the website(www.willselfdestruct.com)	SCREENS HOT 1
E02	82998	I was able to identify the secure anonymous email header that appear on the harassment email sent to the chemistry teacher	It matters because it help me in identifying the secure anonymous email header that appear in the harassment message sent to the chemistry teacher	SCREENS HOT 2
E03	83601	I was able to find the teacher's email address and the harassment message written inside the www.willselfdestruct.com website	It matters because it provide me with the detailed harassment message and also the teacher email address all inside the	SCREENS HOT 3

			website(www.willselfdestruct.com)	
E04	84201	I was able to find the email address of the user that visited the www.willselfdestruct.com website and then wrote the harassment message which was sent to the chemistry teacher	It matters because I was able to obtain the email address if the user that wrote and sent the harassment message to the chemistry teacher	SCREENS HOT 4

Analysis and Discussion

Based on the screenshot provided above, I was able to draw the conclusion that johnny coach with the email address (jcoachj@gmail.com) uses an Apple device with an ip address(192.168.15.4) and MAC Address(00:17:f2:e2:c0:ce) is the person behind the harassment message that was written in the www.willselfdestruct.com website, which he sent directly to his chemistry teacher email address(lilytruckrige@yahoo.com) using the website. Johnny coach is on the chemistry 109 class list provided which prove that he is one of her student

Conclusion

In conclusion this lab has laid a foundation in preparing me on how to analyse pcap capture using wireshark in order to obtain evidences and draw conclusion from them.

Reference

➤ICDFA course Material (SBT-DF204(computer forensic case studies) by COMMANDANT AMINU IDRIS
